

Submission Readiness Assessment: NO-GO

From global heat-allocation bounds to a real waste-heat pathway: a two-scale constraint analysis

DECISION: NO-GO FOR SUBMISSION

1. Executive assessment

The study formulates a global marginal-allocation problem in which heat is conserved and redistributed across space and time to reduce a population-weighted thermal-stress burden, then connects that theoretical screening bound to a measured Frontier waste-heat stream and an ORNL receiving pathway. Technology comparators and transparent techno-economic and operational-emissions accounting are now implemented. A conservative environmental constraint model also excludes unsupported external thermal sinks. A prespecified structural-replication matrix tests year, related reanalysis product, population surface, grid aggregation, and thermal metric choices. Seven finite repeated-dispatch sensitivities now conserve energy at each 10-minute timestep while exposing storage losses, unmet demand, and residual rejection. These additions materially strengthen the project, but they do not establish measured receiving demand, a complete project cost, lifecycle emissions, case-specific cumulative environmental load, or a case-specific integrated engineering design. Submission to Applied Energy should therefore not proceed at this stage.

This decision does not reject the research idea or mathematical framework. It reflects the fixed evidence rule: a gate is closed only by auditable data and reproducible analysis, not by a scenario assumption or narrative qualification.

2. Scientific contribution established to date

1. Formulates conserved heat as spatial relocation, temporal storage, and joint spatiotemporal allocation within one marginal objective.
2. Provides an exact, memory-bounded global one-unit search without constructing an infeasible dense global pair matrix.
3. Diagnoses how distance, land-ocean and cryosphere masks, local capacity, finite heat quantities, and abstract transport or storage penalties alter the mathematical optimum.
4. Shows that low population does not imply a safe heat sink and separates mathematical optima from environmental or engineering recommendations.
5. Integrates public inputs, acquisition ledgers, checksums, analysis code, and artifact generation into a reproducible screening pipeline.
6. Selects a public measured waste-heat case and preserves missing timestamps rather than imputing unsupported annual energy.
7. Separates source capture, heat-pump electricity, storage and delivery losses, auxiliaries, residual rejection, avoided fuel, cost, and operational emissions while preventing COP double counting.
8. Replaces population-only sink safety with source-traced environmental receiving classes, regulatory criteria, physical storage sensitivity, and categorical exclusion of unsupported external discharge.

9. Replicates the central structural findings across prespecified years, related reanalysis products, population surfaces, grid aggregation, and thermal metrics while reporting geographic instability.
10. Implements auditable finite repeated dispatch with source availability, storage state, power limits, losses, auxiliaries, unmet demand, residual rejection, and explicit timestep energy-balance checks.

3. Current evidence

The canonical analysis generated 94 scenarios; relocation dominated no relocation in 93. The representative matched-event spatial bound was 777.8 burden units/GJ. A same-location 18-hour delay yielded 419.3 burden units/GJ, or 53.9% of that event-specific spatial bound.

The matched spatial result had 422 exactly co-optimal candidates and 808 candidates within 1% of the optimum. The representative sink is therefore not unique and must not be interpreted as a policy recommendation.

Phase F evaluated 9 prespecified structural scenarios. Positive constrained spatial value, sink non-uniqueness, pathological unrestricted sinks, positive temporal value, and high-penalty spatial collapse each occurred in 9/9, 9/9, 9/9, 9/9, and 9/9 scenarios, respectively. Source-country labels included Bangladesh, India; maximum source and zero-penalty sink coordinate shifts were 306 km and 1609 km.

The Frontier workbook provides 49,869 valid waste-heat observations totaling 75,573.6 observed MWh-th. 2,691 expected ten-minute timestamps are absent. No missing energy is imputed, and the observed source potential is not presented as delivered useful heat.

Phase D evaluates 3,888 sensitivity cases. For the normalized case-C energy-only calculation using the eGRID SRTV total-output factor, avoided operational emissions are 107.21 kg CO₂e per MWh useful heat. Across the sensitivity grid, avoided operational emissions range from -26.35 to 149.25 kg CO₂e per MWh, and incremental NPC ranges from USD -4,690,918 to USD 6,806,848. Negative outcomes are retained.

The generic full-system comparator has an LCOH of USD 56.51/MWh-th. This is not a Frontier project estimate. Case-specific LCOH, NPC, annual delivery, total residual rejection, and lifecycle CO₂e remain intentionally withheld.

Phase E defines 7 environmental receiving classes and excludes all 2 modeled new external thermal-discharge alternatives. The generic source-backed tank sensitivity contains 9 cases spanning 38.0 to 759.3 MWh-th. This range is not a Frontier storage design; case volume and operating temperatures remain missing.

Phase G evaluates 7 prespecified dispatch sensitivities over 52,560 ten-minute intervals. The zero-demand baseline rejects 75,573.6 MWh-th. The generic-storage lower- and upper-demand sensitivities serve 100.00% and 99.43% of constant reported demand bounds, respectively, while retaining 69,346.2 and 63,501.3 MWh-th of residual rejection. Maximum absolute timestep thermal-balance error is 1.14e-13 MWh. These are sensitivities, not measured Frontier/ORNL operation.

4. Progress since the prior NO-GO assessment

Table 1. Evidence added after the initial audit.

Area	Completed Evidence	Evidence Still Withheld
Real heat case	Measured Frontier source profile and an auditable ORNL receiving pathway selected	Synchronized hourly receiving demand and finite delivery
Technology	Case COP evidence and generic heat-pump, storage, and network comparators	Case capture loss, auxiliaries, route, hydraulics, and balance-of-plant design
Economics and emissions	Energy balance, operational CO ₂ e, CAPEX/OPEX, replacement, discounting, and broad sensitivity	Frontier installed cost, transport CAPEX, case LCOH/NPC, and lifecycle CO ₂ e
Environmental receiving capacity	Source-traced receiving classes, Tennessee thermal criteria, external-sink exclusions, and generic tank sensitivity	Case storage design, route clearance, finite residual load, and any new outfall-specific permit and ecological evidence

Area	Completed Evidence	Evidence Still Withheld
Finite dispatch	Ten-minute source, demand-bound, storage, loss, auxiliary, unmet-demand, and residual-rejection accounting	Measured ORNL demand and case capture, storage, route, outage, and environmental-capacity inputs
Reproducibility	Raw snapshots, checksums, parameter tables, result tables, value ledger, regenerated manuscript package, validation, rendering checks, and tests	Permanent repository DOI and author-controlled submission metadata
Decision evidence	Global-to-real waterfall, six conditional practical modes, post-freeze novelty audit, and adversarial manuscript review	A supported practical endpoint under case-specific inputs

5. Mandatory gate status

Table 2. Evidence-based gate status.

Gate	Status	Blocking Evidence	Required Closure
1. Real source and receiving pathway	PARTIAL—OPEN	Measured source exists; hourly receiving demand and synchronized operation do not	Auditable receiving profile supporting finite dispatch
2. Technology calibration	PARTIAL—OPEN	Generic comparators cannot substitute for case capture, auxiliary, route, and hydraulic measurements	Case-specific design inputs or explicit pathway exclusion
3. Finite operation and dispatch	PARTIAL—OPEN	Repeated sensitivity dispatch is complete, but demand is a constant published bound rather than a measured load trace	Measured receiving demand and case-specific design inputs

Gate	Status	Blocking Evidence	Required Closure
4. Losses, cost, and emissions	PARTIAL—OPEN	Comparator analysis exists; case project cost, residual rejection, and lifecycle boundary remain incomplete	Case LCOH/NPC and supported operational or lifecycle claims
5. Environmental receiving capacity	PARTIAL—OPEN	Sensitivity residual load is quantified, but case storage, route, and permitted receiving capacity remain unresolved	Case-specific cumulative load and receiving-design evidence
6. Structural uncertainty	CLOSED WITH SCOPE LIMITATIONS	Nine prespecified scenarios cover year, related reanalysis product, population surface, grid aggregation, and metric	Retain scope statement: no modern independent high-resolution reanalysis replication

Gate	Status	Blocking Evidence	Required Closure
7. Global-to-real constraint waterfall	CLOSED— NEGATIVE RESULT	Burden-space and physical-energy branches have no supported unit bridge	Maintain separate units and the zero supported endpoint
8. Pareto and mode decision	CLOSED WITH CONDITIONAL MODES	All six evaluated modes depend on incomplete case inputs	Do not identify a supported or preferred practical mode
9. Novelty and journal fit	CLOSED—NO-GO	The applied-energy implementation claim remains unsupported	Acquire integrated case evidence or reconsider journal fit
10. Traceability and reproducibility	CLOSED FOR CURRENT PACKAGE	A permanent repository DOI remains an administrative task	Preserve the frozen ledger and regenerated package
11. Adversarial scientific review	CLOSED—NO-GO RETAINED	No manuscript defect justifies inventing the missing case data	Keep the open evidence gates explicit in all submission files
12. Health-claim control	CLOSED— MAINTAIN	The endpoint must remain a thermal-stress burden proxy	Continue excluding causal health claims

Gate	Status	Blocking Evidence	Required Closure
13. Submission administration	OPEN	Final authorship, declarations, approvals, and repository DOI are unconfirmed	Author confirmation of all submission metadata

6. Case-specific evidence still missing

The Phase D and Phase E parameter registries preserve the following items as missing-required rather than substituting generic values:

- Case-specific balance-of-plant auxiliary electricity: Do not add the generic 1% value to a COP that may already include unit auxiliaries.
- Frontier source-side heat capture efficiency: No measured source heat-exchanger loss is available.
- Frontier full installed project cost and price year: ORNL reports equipment-only cost with incomplete price-year and balance-of-plant scope.
- Frontier-specific hot-water network CAPEX: No route, hydraulics, or project quote is available.
- Equipment, construction, refrigerant, and replacement lifecycle emissions: Operational factors do not support a cradle-to-grave LCA.
- Hourly ORNL receiving heat demand: Annual dispatch and total residual rejection cannot be calculated.

- Annual residual Frontier heat rejection after dispatch: Requires synchronized source, demand, storage, and outage dispatch.
- candidate receiving-water flow: Required to convert temperature criteria to a hydraulic heat-load bound.
- candidate receiving-water upstream temperature: Required to evaluate the 30.5 deg C absolute criterion.
- approved thermal mixing-zone definition: No candidate outfall or approved mixing zone exists.
- aquatic biological and cumulative-impact assessment: Required before any practical thermal-discharge recommendation.
- case-specific tank volume: Required to fix the practical storage capacity.
- case-specific storage temperature difference: ORNL return temperature and storage design remain unavailable.
- route and footprint environmental clearance: No hot-water route or project footprint has been defined.
- annual residual rejected heat: Requires synchronized source, demand, storage, and outage dispatch.

7. Minimum conditions for a GO decision

- Obtain an auditable receiving-demand profile and replace the constant demand-bound sensitivities with case-specific finite dispatch, including supported capture, storage, route, outage, and receiving-capacity inputs.

- Resolve or explicitly exclude unsupported source-capture, auxiliary, network, hydraulic, installed-cost, and lifecycle-emissions terms.
- Supply the case storage, route, and finite residual-load evidence required by the environmental constraint model; retain categorical exclusion of any unsupported external sink.
- Confirm authorship, affiliation, funding, conflicts, originality, author approval, and a permanent repository identifier.
- Repeat the frozen-value, adversarial-review, and clean-build checks after any new evidence changes the practical endpoint.

8. Claims permitted at the current stage

Table 3. Current claim boundary.

Permitted	Not Permitted
A theoretical conserved-heat allocation screening bound and pathology diagnosis	A practical recommendation to move heat to the selected sink
Measured Frontier source potential over valid observations	Measured complete-year Frontier/ORNL delivery from sensitivity dispatch
Finite repeated-dispatch sensitivities under constant published demand bounds and generic storage assumptions	Calibrated or observed annual Frontier/ORNL operation
Normalized operational-emissions and generic cost comparators	Frontier project LCOH, NPC, payback, or lifecycle CO ₂ e

Permitted	Not Permitted
Population-weighted thermal-stress burden	Mortality, morbidity, or causal health benefit

9. Recommended next decision

Do not submit the package to Applied Energy at the current evidence state. The remaining paths are to obtain synchronized demand and an integrated route, capture, auxiliary, storage, cost, lifecycle, and environmental design, or to retain the transparent negative result and reconsider the journal and article type.

10. Conclusion

Final decision: NO-GO. The mathematical framework, real-case source evidence, technology comparators, Phase D accounting, environmental exclusions, and Phase F structural replication, Phase G dispatch sensitivities, constraint waterfall, mode decision, frozen ledger, and adversarial review constitute meaningful progress. However, measured finite delivery, case-specific engineering and economics, case-specific cumulative environmental load, and submission administration remain unresolved. The decision should not change to GO until the mandatory gates are closed with auditable evidence.